

MINI CNC PLOTTER

Mini Project Report Submitted in partial fulfilment of the requirements for
the degree of Bachelor of Technology from Maulana Abul Kalam Azad
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By

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Publications and Acronyms

No academic research papers, patents, or conference presentations have been published from the research work and design findings presented in this mini project report so far. The system has been designed and implemented primarily as an educational prototype to fulfill the academic requirements of the Bachelor of Technology degree in Electronics and Communication Engineering.

The following table lists the principal abbreviations, symbols, acronyms, and technical terms used in this mini project report, along with their definitions.

Acronym / Symbol	Description / Meaning
ECE	Electronics and Communication Engineering
CNC	Computer Numerical Control
GRBL	Embedded Open-Source G-code Parser Firmware
SVG	Scalable Vector Graphics
PWM	Pulse Width Modulation (used to control motor speed)
UGS	Universal Gcode Sender
UART	Universal Asynchronous Receiver-Transmitter
GND	Electrical Common Ground

Table 1: List of Acronyms and Symbols

Contents

Acknowledgement	1
Publications and Acronyms	2
Lists of Figures and Tables	4
Abstract	5
1 Introduction	6
1.1 Project Overview	6
1.2 Review of Previous Works.....	6
1.3 Objective and Motivation	6
1.4 Scope of Present Work	7
2 Hardware Architecture and Component Details	8
2.1 Microcontrollers	8
2.2 Stepper Motors	8
2.3 Stepper Motor Driver Modules.....	8
2.4 3D Printed Mechanical Structure.....	9
2.5 Connecting Wires	10
2.6 External Power Supply (5V–9V DC).....	10
2.7 Connecting Wires	10
2.8 External Power Supply (5V–9V DC).....	10
2.9 Pen or Marker	10
3 System Design and Circuit Integration	11
3.1 System Architecture.....	11
3.2 Embedded Controller Interface.....	11
3.3 Power and Driver Stage Integration	11
3.4 Multi-Axis Actuation.....	12
4 Software Implementation and Control Algorithms	13

5	Conclusions and Future Scope	14
5.1	Performance Evaluation.....	14
5.2	Conclusions	14
5.3	Future Scope.....	14
	References	15

List of Figures

I.	Figure 1 – Arduino Uno.....	8
II.	Figure 2 - stepper motor (28BYJ-48) and motor driver ULN2003	9
III.	Figure 3 –3D Printed Mechanical Structures	9
IV.	Figure 4 - Circuit connections and diagram.....	12
V.	Figure 5 – GRBL Library for Arduino based Mini CNC Plotter	

List of Tables

1	List of Acronyms and Symbols	2
1.1	Project System Specifications.....	7

Abstract

Abstract This project presents the design, development, and implementation of a low-cost, compact, three-axis Computer Numerical Control (CNC) plotter machine. The system is engineered to bridge the gap between digital vector designs and automated physical sketching on a Cartesian plane. The hardware architecture utilizes a robust framework of 3D-printed structural parts driven by three hobby stepper motors and integrated H-bridge driver modules, ensuring electrical isolation and precise phase control.

At the core of the system, an Arduino microcontroller running open-source GRBL firmware serves as the real-time motion controller. The digital software toolchain transforms raw raster images into scalable vector graphics (SVG) using Inkscape, generates specialized toolpaths (G-code) via JSCut, and streams coordinate packets sequentially over a serial interface using Universal Gcode Sender (UGS). The firmware parses these instructions into synchronized digital step and direction pulse trains to achieve high-accuracy spatial mapping within a defined $7\text{cm} \times 7\text{cm}$ workspace.

This project successfully demonstrates key Electrical and Electronics Engineering principles, including embedded systems firmware implementation, power management isolation between logic and inductive loads, and automated open-loop control systems. The resulting prototype offers a scalable, cost-effective foundation for advanced mechatronic applications such as automated PCB prototyping and laser engraving

Chapter 1 : Introduction

1.1 Project Overview

Computer Numerical Control (CNC) technology has traditionally been locked behind expensive, large-scale industrial machinery. However, the rise of modern open-source microcontrollers and desktop manufacturing tools like 3D printing has democratized rapid prototyping. A CNC plotter is a miniature mechatronic application that translates computer-generated graphics into precise physical movements over X, Y, and Z spatial coordinates. Studying these systems offers valuable insights into real-time embedded systems engineering.

1.2 Review of Previous Works

Earlier variations of desktop plotters relied primarily on recycled CD/DVD drive stepper mechanisms. While cost-effective, those implementations drastically limited the work envelope and mechanical rigidity. Subsequent research integrated specialized CNC shields (like the CNC Shield V3) paired with high-current A4988 drivers. Recent shifts focus heavily on making these structures completely customizable using CAD modeling and 3D printing, alongside open-source optimization frameworks like GRBL to replace expensive commercial motion-control software.

1.3 Objective and Motivation

The core objective of this mini-project is to build an autonomous, budget-friendly 3-axis mini CNC plotter utilizing basic electronics laboratory equipment and 3D-printed structural frames. The motivation stems from understanding structural automation—specifically learning how an 8-bit microcontroller can compute real-time coordinate math, handle high-current inductive isolation, and convert complex graphical shapes into highly synchronized pulse trains.

1.4 Scope of Present Work

The scope of this work encompasses the electromechanical assembly, firmware calibration, and signal optimization of a 3-axis CNC platform limited to a $7\text{ cm} \times 7\text{ cm}$ workspace. It focuses on establishing a full digital-to-analog hardware pipeline: converting vector graphics into discrete alphanumeric G-code, transmitting data via a USB-UART serial interface, and using an 8-bit microcontroller to parse inputs into synchronized PWM step/direction pulse trains. Additionally, the work isolates low-power logic control from high-current inductive motor rails to ensure system stability and eliminate electromagnetic interference (EMI).

To compare these systems, Table 1.1 outlines the technical specs of the ESP32 design.

Feature	CNC Plotter Hardware
Microcontrollers	Arduino Uno R3(8-bit AVR RISC architecture)
Actuators	Three Hobby Stepper Motors (28BYJ-48 or equivalent)
Motor Drivers	Three Stepper Driver Modules (ULN2003)
Firmware Backbone	Open-source GRBL (High-performance G-code parser)
Communication Protocol	USB-to-UART Serial Interface (via Universal Gcode Sender)
Working Envelope	$7\text{ cm} \times 7\text{ cm}$ Cartesian coordinate space
Power Architecture	Isolated Dual Rails: 5V Logic (USB) + 5V – 9V Inductive Motor Drive

Table 1.1: Project System Specifications

Chapter 2 : Hardware Architecture and Component Details

2.1 Microcontrollers

This project uses the Arduino Uno. It serves as the main control unit of the mini CNC plotter. It receives G-code instructions from a computer through a USB connection and processes them to control the movement of the X, Y, and Z axes. It generates the necessary control signals for the stepper motor drivers and coordinates the overall operation of the machine. The Arduino is widely used because it is inexpensive, easy to program, and compatible with GRBL-based CNC firmware.



Fig1. Arduino UNO

2.2 Stepper Motors

Three stepper motors (28BYJ-48) are used to provide precise and controlled movement of the plotter. Two motors control the X-axis and Y-axis, allowing the pen to move horizontally across the paper, while the third motor controls the Z-axis by lifting or lowering the pen. Stepper motors rotate in fixed angular steps, making them ideal for applications that require high positioning accuracy.

2.3 Stepper Motor Driver Modules

Stepper motor driver modules ($3 \times \text{ULN2003}$) act as the interface between the Arduino and the stepper motors. Since the Arduino cannot supply enough current to drive the motors directly, these driver modules amplify the control signals and provide the required power. They also allow precise control of motor speed, direction, and stepping sequence, ensuring smooth and accurate movement.

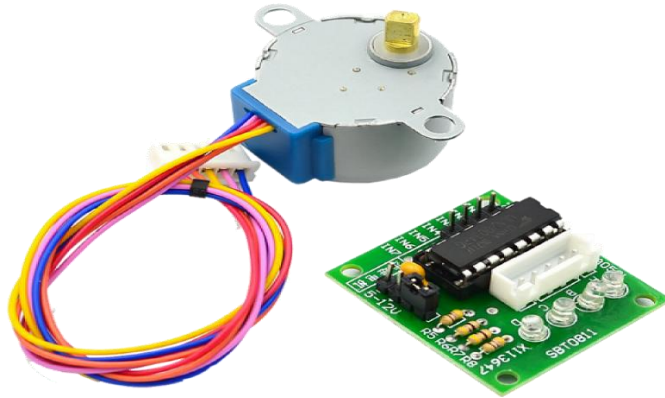


Fig 2. - stepper motor (28BYJ-48) and motor driver ULN2003

2.4 3D Printed Mechanical Structure

The mechanical frame of the CNC plotter consists of 3D-printed parts, including the base frame, axis supports, carriage, and pen holder. These components provide a lightweight yet sturdy structure that holds all the hardware in proper alignment. The use of 3D printing allows easy customization, low manufacturing cost, and accurate fitting of components.



Fig 3. 3D Printed Mechanical Structures

2.5 Connecting Wires

Jumper wires and connecting cables are used to establish electrical connections between the Arduino, motor drivers, stepper motors, and power supply. Proper wiring ensures reliable communication and power distribution throughout the system while minimizing electrical faults and signal interference.

2.6 External Power Supply (5V–9V DC)

An external DC power supply is used to provide sufficient power to the stepper motors and motor driver modules. Since the motors require more current than the Arduino can provide, a separate power source ensures stable operation and prevents overheating or voltage drops. A regulated 5V to 9V supply is commonly used for small CNC plotters.

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2.9 Pen or Marker

A pen or marker is mounted on the pen holder to draw images, text, or designs on paper. The pen moves according to the commands received from the Arduino, while the Z-axis motor lifts or lowers it during plotting.

Chapter 3 : System Design and Circuit Integration

3.1 System Architecture

The system design of the Mini CNC Plotter is built on a modular mechatronic framework. It is divided into three primary functional domains: the Host Controller (Software), the Embedded Controller (Logic), and the Power and Actuation Stage (Hardware).

3.2 Embedded Controller Interface

The central processing unit is an Arduino Uno board equipped with an ATmega328P 8-bit AVR microcontroller.

Firmware Execution: The Arduino is flashed with GRBL firmware. It operates as a real-time G-code parser and state machine.

I/O Mapping: Digital output pins of the microcontroller are configured to generate discrete, high-frequency Step (pulse) and Direction (high/low logic level) signals. These signals are mapped directly to the input control pins of the respective stepper drivers.

UART Communication: The Arduino communicates with the host machine via an on-board USB-to-UART bridge (e.g., CH340G or ATmega16U2) operating at a baud rate of 115200 bps to receive coordinate streams sequentially. They are configured as shown in Tables 3.2 and 3.3.

3.3 Power and Driver Stage Integration

A critical design requirement in mechatronic systems is electrical isolation between low-power logical circuits and high-power inductive loads.

Driver Topology: The system integrates three independent stepper motor driver modules (such as ULN2003 transistor). These modules utilize internal H-bridge configurations to handle the sequential phase-switching required by the stepper motor coils.

Power Decoupling: To prevent high-current inductive spikes from causing transient voltage drops—which lead to microcontroller resets—the system utilizes a dual-rail power architecture:

Logic Rail (V_{CC}): Supplied with stable 5 V

directly from the USB port to power the microcontroller. Motor Rail (V_M): Driven by an external 5 V – 9 V DC power supply connected directly to the motor drivers to safely sink the current demands of the motors.

3.4 Multi-Axis Actuation

X and Y Axis Control: Two stepper motors translate rotation into linear coordinate movement. Each step signal pulsed by the Arduino rotates the motor by its fundamental step angle (e.g., $5.625^\circ/64$ reduction), moving the axis along the Cartesian grid with high precision. **Z-Axis Control:** A third stepper motor acts as a binary pen-lift mechanism. When a drawing path is active (e.g., G1 coordinate command), the driver lowers the pen; when a non-drawing translational move is commanded (e.g., G0 fast travel), the motor lifts the pen to prevent trace artifacts on the workspace.

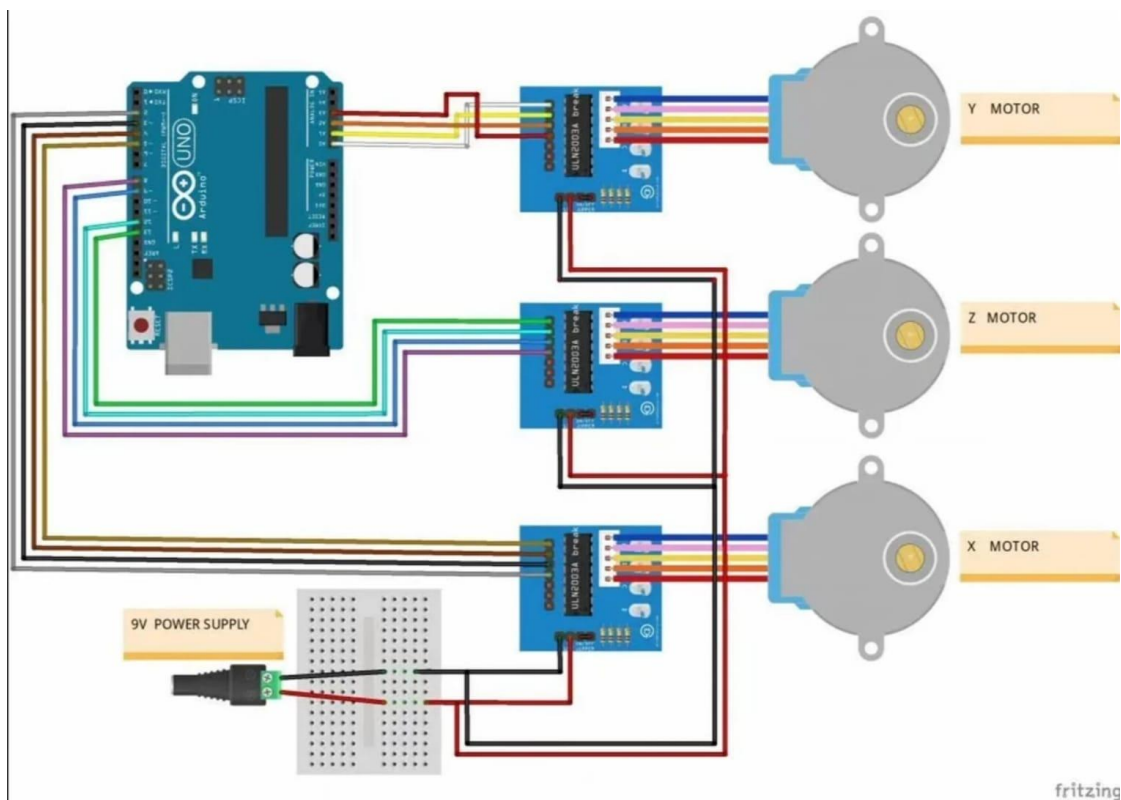


Fig 4. Circuit connections and diagram

Chapter 4 : Software Implementation and GRBL Library

GRBL is a free, open-source firmware that enables an Arduino Uno (ATmega328P) to function as a CNC (Computer Numerical Control) machine controller. It interprets G-code commands sent from a computer and converts them into precise step and direction signals for stepper motor drivers, allowing accurate control of CNC machines, laser engravers, plotters, and other automated devices.

GRBL is designed to be lightweight, efficient, and reliable, making it one of the most widely used firmware solutions for DIY and low-cost CNC systems. It supports communication through a USB serial interface and is compatible with many G-code sender software applications.

Uses of GRBL Library

1. CNC Machine Control: Controls CNC milling, drilling, and cutting machines accurately.
2. Laser Engraving and Cutting: Operates laser engravers and cutters for engraving designs on wood, acrylic, leather, and other materials.
3. Plotters and Drawing Machines: Controls XY plotters for drawing diagrams, artwork, and PCB layouts.
4. 3-Axis Motion Control: Provides synchronized movement of X, Y, and Z axes for automated machines.

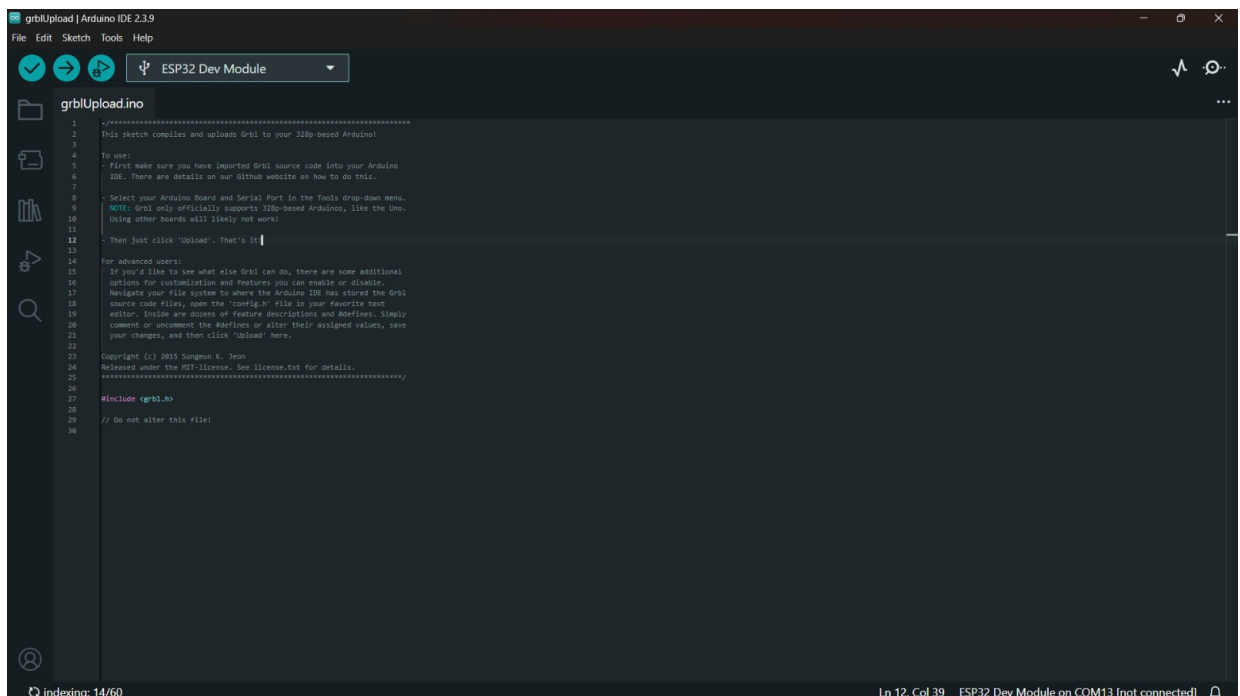


Fig 5. GRBL Library for Arduino based Mini CNC Plotter

Chapter 5 : Conclusions and Future Scope

5.1 Performance Evaluation

The prototype was subjected to operational testing to evaluate spatial accuracy, system latency, and thermal stability: **Dimensional Accuracy:** The plotter successfully executed complex vector designs within the 7 cm × 7 cm workspace with minimal geometric distortion. **Mechanical Backlash:** Rigid 3D-printed axis guides minimized mechanical play, ensuring a repeatable trace width matching the 0.3 mm pen diameter. **Thermal and Power Isolation:** Decoupling the 5V logic power rail from the external 5V – 9V stepper driver rails eliminated voltage drops and electromagnetic interference (EMI) during continuous plotting cycles.

5.2 Conclusions

This mini-project successfully validates the design of a low-cost, precise mini CNC plotter utilizing an open-source hardware and software ecosystem. **Successful Integration:** This project successfully demonstrates the integration of an 8-bit AVR microcontroller running GRBL firmware with discrete H-bridge stepper drivers to achieve precise open-loop motion control. **Robust Data Pipeline:** The design successfully validates a complete digital-to-analog workflow, processing digital vector images into physical coordinate motion. **Cost-Effectiveness:** The prototype represents a low-cost, highly accessible educational platform for studying mechatronics, embedded systems, and automated spatial mapping.

5.3 Future Scope

Laser Engraving Module: The binary Z-axis pen lift mechanism can be swapped with a low-power, pulse-modulated laser diode to convert the plotter into a desktop laser engraver. **Automated PCB Prototyping:** The platform can be adapted to draw direct etch-resist traces on copper-clad boards using permanent markers before chemical etching. **Closed-Loop Feedback:** Integrating optical rotary encoders on the stepper shafts will transition the system from open-loop to closed-loop control, drastically improving error correction and overall tracing resolution.

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